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Measures of Central Tendency

Measures of Central Tendency (The Mean)

4.1 Introduction

- One of the most important objectives of statistical analysis is to get *one single value* that describes the characteristic of the entire mass of data.
- Such a value is called the *central value* or an *average* or the expected value of the variable.

Measures of Central Tendency (The Mean)

- The goal in measuring central tendency is to describe a group of individuals (more accurately, their scores) with a single measurement.
- Ideally, the value we use to describe the group will be the single value that is *most representative* of all the individuals.

Measures of Central Tendency (The Mean)

- Statisticians sometimes use the expression “*number crunching*” to illustrate this aspect of data description.
- That is, we take a distribution consisting of many scores and “*crunch*” them down to a single value that describes them all.

Measures of Central Tendency (The Mean)

- *Average is defined as attempt to find a single figure to describe whole of figures.*
- The most common measures of central tendency are the *mean*, the *median* and the *mode*.

Measures of Central Tendency (The Mean)

We divide MCT in to two measure parts:

☐ *Mathematical Measures of central Tendency*

Here we include *simple arithmetic mean, weighted arithmetic mean, geometric mean and harmonic mean.*

Measures of Central Tendency (The Mean)

- *Positional Measures of central Tendency*

And here we include the *median, quartiles, deciles and percentiles*.

- The *mode* is the type of MCT which focuses on counting the most frequently occurred data.

Measures of Central Tendency (The Mean)

Types of Arithmetic mean

Arithmetic mean is of the following two types:

***(i).* Simple Arithmetic Mean:**

In simple arithmetic mean, all items of a series are given equal importance.

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4.2 (i) The Mean (Simple Arithmetic Mean)

Population and Sample Mean

- Measures of central tendency are defined for a *population data*, denoted by μ (read as “*miw*” and for a *sample data*, denoted by $\bar{x}$ (read as *x* bar).

Measures of Central Tendency (The Mean)

“Arithmetic mean is defined as sum of the different values of an item of a series divided by total number of items.”

Thus Population arithmetic mean is given by:

$$\mu = \frac{\text{Sum of all values in the population}}{\text{Number of values in the population}}$$

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and the sample arithmetic mean is given by:

$$\bar{x} = \frac{\text{Sum of all values in the sample}}{\text{Number of values in the sample}}$$

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Definition: The arithmetic mean (or simply the mean) of set of n observations $x_1, x_2, x_3, \dots, x_n$, denoted by $\bar{x}$ is defined as

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}.$$

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Measures of Central Tendency (The Mean)

Example 1:- Find the arithmetic mean of

2, 9, 3, 7, 6, 3, 2, 3, 6, 9

Solution:- $\bar{x} = \frac{2 + 9 + 3 + 7 + 6 + 3 + 2 + 3 + 6 + 9}{10} = \frac{50}{10} =$

5.

[?] If the numbers x_1 occurs f_1 times, x_2 occurs f_2 times, x_3 occurs f_3 times, . . . , x_k occurs f_k times, then

$$\bar{x} = \frac{x_1 f_1 + x_2 f_2 + x_3 f_3 + \dots + x_k f_k}{f_1 + f_2 + f_3 + \dots + f_n} = \frac{\sum_{i=1}^n x_i f_i}{n}.$$

Measures of Central Tendency (The Mean)

Example 1:- Considering example 1 and formula (2), find the arithmetic mean of the given data.

Solution:- Given 2, 9, 3, 7, 6, 3, 2, 3, 6, 9.

Putting these values with respect to their frequencies in a table, we obtain:

Value	2	3	6	7	9
Frequency	2	3	2	1	2

This type of table is known as ungrouped FD.

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$$\begin{aligned}\text{Thus } \frac{\sum_{i=1}^k x_i f_i}{\sum_{i=1}^k f_i} &= \frac{\sum_{i=1}^5 x_i f_i}{\sum_{i=1}^5 f_i} \\&= \\&\frac{x_1 f_1 + x_2 f_2 + x_3 f_3 + x_4 f_4 + x_5 f_5}{f_1 + f_2 + f_3 + f_4 + f_5} \\&= \frac{2(2) + 3(3) + 6(2) + 7(1) + 9(2)}{2 + 3 + 2 + 1 + 2} = \\&\frac{4 + 9 + 12 + 7 + 18}{10}\end{aligned}$$

Measures of Central Tendency (The Mean)

Example2:- A sample of five executive bankers received the following bonus in 2015 Ethiopian calendar (in thousands of birr).

47.3, 38.6, 55.0, 25.8, 40.3. Find the average bonus received by the bankers.

Measures of Central Tendency (The Mean)

$$\bar{x} =$$

$$47.3 + 38.6 +$$

$$207$$

Measures of Central Tendency (The Mean)

(ii) **Weighted Arithmetic Mean:**

Under weighted A.M. different items of the series are assigned weights according to their relative importance.

i.e. weighted mean is the mean of a set of values where each value or measurement has a *different* *weight* or *degree of importance*.

Measures of Central Tendency (The Mean)

It is defined by the formula:

$$\begin{aligned}\bar{x} &= \frac{x_1 w_1 + x_2 w_2 + \dots + x_k w_k}{w_1 + w_2 + w_3 + \dots + w_n} \\ &= \frac{\sum_{i=1}^n x_i w_i}{\sum_{i=1}^n w_i} \dots (3)\end{aligned}$$

Measures of Central Tendency (The Mean)

Example3:- A student scored **A** in a 3 credit hours course, **B** in a 5 credit hours course and **D** in a 4 credit hours course. This student is going to apply for a competition which requires a minimum GPA of 2.5. Suppose **A** has 4 points, **B** has 3 points and **D** has 1 point. Check whether a student is going to succeed to fulfil the requirement or not.

Measures of Central Tendency (The Mean)

Solution:- Let's form the following table to make the given data more clear.

Grades	A	B	D
Points(x_i)	4	3	1
Weights(w_i)	3	5	4

Now as the weighted mean is defined as

$$\bar{x} = \frac{x_1 w_1 + x_2 w_2 + \dots + x_k w_k}{w_1 + w_2 + \dots + w_k} = \frac{\sum_{i=1}^n x_i w_i}{\sum_{i=1}^n w_i} \text{ we}$$

can substitute the values of x_i and w_i to find the value of the AM.

Measures of Central Tendency (The Mean)

$$\begin{aligned}\text{Thus } \bar{x} &= \frac{x_1 w_1 + x_2 w_2 + x_3 w_3}{w_1 + w_2 + w_3} = \\ &= \frac{4(3) + 3(5) + 1(4)}{3 + 5 + 4} \\ &= \frac{12 + 15 + 4}{12} = \frac{31}{12} \approx 2.58.\end{aligned}$$

So we can conclude that a student fulfills the requirement since his GPA > 2.5 .

Measures of Central Tendency (The Mean)

4.2.1 *Properties of the arithmetic mean*

Suppose we have N observations:

$x_1, x_2, x_3, \dots, x_N$ whose mean is given by $\bar{x}$ and let k be any constant number. Then

(i) the arithmetic mean of

a) $x_1 + k, x_2 + k, x_3 + k, \dots, x_N + k$ is $\bar{x} + k$

b) $x_1 - k, x_2 - k, x_3 - k, \dots, x_N - k$ is $\bar{x} - k$

c) $x_1 \times k, x_2 \times k, x_3 \times k, \dots, x_N \times k$ is $\bar{x} \times k$

Measures of Central Tendency (The Mean)

Example 1:- Consider the data 3, 5, 6, 7, 9.

We can see that $\bar{x} = \frac{3 + 5 + 6 + 7 + 9}{5} = \frac{30}{5} = 6$.

(i) Add **2** on each of the values and obtain

5, 7, 8, 9, 11

Now $\bar{x}_{New} = \frac{5 + 7 + 8 + 9 + 11}{5} = \frac{40}{5} = 8 = \bar{x}_{old} + \mathbf{2}$

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(ii) Subtract **1** on each of the values and obtain

2, 4, 5, 6, 8

$$\text{Thus } \bar{x}_{New} = \frac{2 + 4 + 5 + 6 + 8}{5} = \frac{25}{5} = 5 = \bar{x}_{old} - \mathbf{1}$$

(ii) Multiply each value by **3** and obtain

9, 15, 18, 21, 27

$$\text{Here } \bar{x}_{New} = \frac{9 + 15 + 18 + 21 + 27}{5} = \frac{90}{5} = 18 = \bar{x}_{old} \times \mathbf{3}$$

Measures of Central Tendency (The Mean)

Example2:- The average selling price of items in Abadir Supermarket is 436 birr. If the discount of 5% is made on each of the items in the Super Market for the Arafa Holiday, find the average selling price of the items after the discount.

Measures of Central Tendency (The Mean)

2. Since $\bar{x} = \frac{\sum_{i=1}^N x_i}{N}$, it follows that

$$\sum_{i=1}^N x_i = \bar{x} \times N.$$

We use this property in *correcting* and *combining* the arithmetic mean.

3. The sum of the deviation of the given observations from the mean is always zero.

$$\text{i.e. } \sum_{i=1}^N (x_i - \bar{x}) = 0.$$

Measures of Central Tendency (The Mean)

Example 3:- Consider the data given in example 1.

i.e. 3, 5, 6, 7, 9.

We know that $\bar{x} = 6$.

Now take the difference of $x_i - \bar{x}$.

Measures of Central Tendency (The Mean)

$$\text{Thus } x_1 - \bar{x} = 3 - 6 = -3$$

$$x_2 - \bar{x} = 5 - 6 = -1$$

$$x_3 - \bar{x} = 6 - 6 = 0$$

$$x_4 - \bar{x} = 7 - 6 = 1$$

$$x_5 - \bar{x} = 9 - 6 = 3$$

As we can see

$$\sum_{i=1}^N (x_i - \bar{x}) = (-3) + (-1) + 0 + 1 + 3 = 0.$$

Measures of Central Tendency (The Mean)

- *The AM for a GFD*

The arithmetic mean of a grouped frequency distribution is computed using the following formula:

$$\bar{x} = \frac{\sum_{i=1}^k x_i f_i}{N}, \text{ where}$$

x_i - class mark of the i^{th} class

f_i - frequency of the i^{th} class

k – the number of classes of the GFD

N - the number of observations in the GFD

Any Questions



Any Questions

